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CRUCIBLE DE-WETTING DURING BRIDGMAN GROWTH OF SEMICONDUCTORS IN MICROGRAVITY

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After a literature survey and observations made during a space experiment, the phenomenon of crucible de-wetting by the crystal during Bridgman solidification in microgravity is explained by a model involving composite wetting of the crucible by the liquid, crystal angle of growth and interface advance. A ground experiment was run in order to validate this model which also explains why a crystal detaches from the crucible surface when a sand blasted crucible is used in Bridgman solidification on the ground. It is shown that de-wetting leads to enhanced quality of the crystal produced and that capillary-induced convection effects are not to be feared in this case. Consequently, it is highly advisable to use rough-surface crucibles for crystal growth both in microgravity and on the ground.

1. Introduction

In order to improve the electronic properties of semiconductor crystals produced from a liquid phase, it is essential to reduce the number of defects caused by convection-induced movements within the liquid, such as striations or longitudinal or radial differences in chemical composition. Crystal growth experiments performed in microgravity have shown that, as forecast by theoretical considerations, homogeneous solutal distribution profiles can be obtained without convection [1–4]. However, if the liquid has free surfaces, convective currents may be produced at the surface as a result of surface tension variations caused by temperature differences (thermocapillary convection or Marangoni effect) or by chemical composition differences. These current effects are transferred to the entire liquid by viscosity, thereby considerably disturbing the diffusion-based growth conditions [5,6]. It is important to study the free surface appearance mechanisms so as to control and quantify their effect on solidification.

Table 1 gives an overview of observations made concerning the spontaneous appearance of free surfaces during metallurgy experiments in microgravity. The observed phenomena can be placed in three different categories:

(1) *Free volumes and bubbles.* These may be produced intentionally (for example to compensate for sample volume variations) or unintentionally (sample degassing on fusing, filling defect, liquid separation into several zones). Sen and Wilcox have studied this phenomenon [27,28] and have showed that a variety of configurations are energetically stable, especially the presence of bubbles attached to the crucible or the division of the liquid into several independent volumes. A large free surface area is thus often created (several millimetres in length) and this can very easily give rise to intense convective mixing. This is particularly the case of bubbles [22,25,26] which are frequently observed (table 1 quotes just a small number of cases), in as much that the absence of gravity does not allow liquid and gaseous phases to be separated.

(2) *Necking.* This phenomenon is characterised by a considerable reduction in crystal diameter at the start of growth. Zemskov explained this phenomenon [18] by assuming that, in this case, the sum of the growth angle, α (also called "angle of solid wetting by its liquid" by certain authors

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Table 1
Review of free-surface occurrence during microgravity solidification experiments

Experiment	Sample (+dopant) [a]	Crucible	Atmosphere [b]	Free surface	Convective mixing	References
Yue and Voltmer Skylab.	$Ge+8\times10^{16}$ Ga $Ge+4\times10^{14}$ Sb $Ge+2\times10^{15}$ B	Graphite	$<10^{-4}$ Torr	Necking, de-wetting?	$\mu g<1g$	[7,8]
Witt et al. Skylab 3	$InSb+10^{18}$ Te	SiO_2	10^{-7} Torr BS	De-wetting	No	[3,9,10]
Witt et al. Skylab 4	$InSb+10^{18}$ Te	SiO_2	10^{-7} Torr BS	Necking, de-wetting	No	[3,10]
Witt et al. Skylab 3	$InSb+10^{20}$ Sn InSb	SiO_2	10^{-7} Torr BS	(Little bubbles)	?	[3]
Ukanwa Skylab 3	Zn + Zn 65	Graphite	Argon	De-wetting	Negligible	[11]
Yee et al. Skylab 3-4	$Ga_xIn_{1-x}Sb$ $x=10\%, 30\%, 50\%$	SiO_2 graphitized	Helium 10 Torr	Necking	?	[12–14]
Hasemeyer et al. Skylab 3-4	Al_2–Cu eutectic	Graphite	?	Necking	?	[15]
Witt et al. ASTP	$Ge+10^{19}$ Ga	Machined SiO_2	Argon 40 Torr	De-wetting	No	[4,16]
Zemskov et al. ASTP	$Ge+1\%$ Si $+10^{17}$ Sb	SiO_2 graphitized	10^{-4} Torr BS	De-wetting	No	[17]
Zemskov et al. Salyut 6	$InSb+1.3\times10^{19}$ Te	SiO_2 graphitized	?	Necking, de-wetting, bubbles	Marangoni	[1,18]
Markov et al. Salyut 6	$Ge+10^{17}$ Ga	SiO_2 graphitized	?	De-wetting	No	[19]
Kashimov et al. Salyut 6	InSb $InSb+10^{18}$ Te	SiO_2 graphitized	10^{-7} Torr BS	De-wetting	No	[2]
Lendvay et al. Salyut 6	GaSb	SiO_2 graphitized	10^{-6} mbar BS	De-wetting	?	[20,21]
Regel and Nghi Salyut 6	$Bi_2(Te, Se)_3$	SiO_2	?	Bubbles	Marangoni in bubbles	[22,23]
Rodot and Tottereau FSLP	$PbTe+10^{18}$ Ag	SiO_2	Argon	Bubbles	?	[24]
Duffar et al. D-1	$InSb+9\times10^{19}$ Sn	Machined SiO_2	10^{-6} mbar BS $<10^{-4}$ at opening	Bubbles, partial de-wetting	Marangoni in bubbles	[25,26]

[a] 8×10^{16} Ga means: Ga dopant at 8×10^{16} cm^{-3} level.
[b] BS: before sealing.

[29,30]), and the wetting angle, θ, is greater than 180° (fig. 1) thereby necessarily drawing the solid away from the crucible. This explanation is confirmed by the fact that this phenomenon appears only in the case of semiconductors (α ranging from less than 10° to almost 50°, table 2) in a

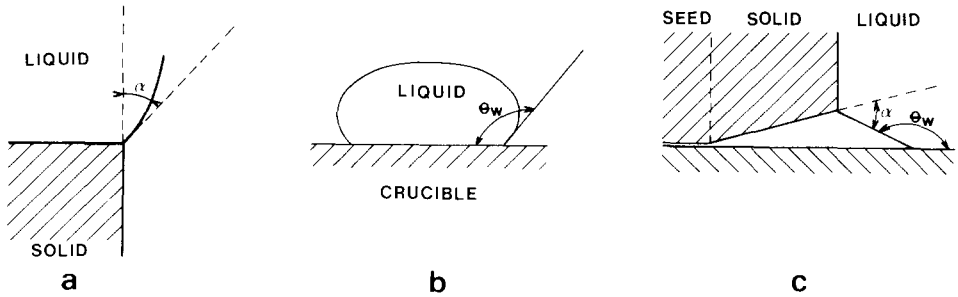

Fig. 1. Basic concepts of wetting behaviour: (a) growth angle α; (b) wetting angle θ_w; (c) necking (after Zemskov et al. [18]).

graphite crucible ($\theta \geq 140°$, table 3). If the necking diameter is small, the liquid surface can then become quite considerable, promoting the development of convective movements of increasing disruptiveness the closer they are to the solidification interface [18]. Tables 2, 3 and 4 give the values of growth angle and wetting angle for the samples and crucibles in table 1.

(3) *De-wetting.* This common phenomenon in microgravity results in a crystal surface state without any apparent relation to the surface state of the crucible (generally rough): the smooth or faceted surfaces are bounded by ridges several dozen microns thick.

To the authors' knowledge, while the creation of free surfaces belonging to the first two categories has been studied in detail and is now well understood, the same cannot be said for the de-wetting phenomenon:

– The phenomenon is particularly prevalent among semiconductors. In the table, only two experiments concern metals and alloys [11,15] and are not very conclusive. Ahlborn and Lohberg [40], in the case of lead, explain this type of observation by shrinkage. This explanation cannot be used in the case of semiconductors as they expand on solidifying. Growth angles (Table 2) are virtually zero for metals and high for semiconductors; it is therefore possible that growth angle plays a role in this phenomenon.

– The nature of the semiconductor, the presence and concentration of possible doping agents and the type of ambient atmosphere would not appear

Table 2
Growth angles of a number of metals and alloys

Product	Growth angle (deg)	Ref.
Cu	0	[31]
Ga	0	[32]
InSb	30	[18]
	25	[33]
	20–30	[34]
Si	10	[34]
	11	[29]
Ge	12	[34]
	8	[29]
	7	[32]
Ga + 3% As/GaAs	40	[35]
Ga–In–As/GaAs	32	[35]
InP	48	[32]

Table 3
Wetting angle of various metals and alloys on graphite at their melting point

Product	Wetting angle (deg)	Ref.
InSb	165	[18]
In	141	[36]
Sb	140	[36]
Ge	139	[36]
Zn	150	[36]
Ga	137	[36]
Pb	138	[36]
Cu	140	[36]
Al	157	[36]
Si	15–0	[36]
Ga + 3% As	140–153	[35]
InP	144	[37]
In	146	[37]
GaInAs	148	[37]

Table 4
Wetting angle of various metals on silica at their melting point (after refs. [38,39])

Product	Wetting angle (deg)
Al	90
Ge	106
In	128
Sb	111
Cu	128
Ga	119
Pb	120
Zn	100

to have any bearing on the appearance of de-wetting.

– Crucibles are made of silica or graphite. Tables 3 and 4 show poor wetting of these materials by semiconductors (or their constituents).

Growth of semiconductors in microgravity without de-wetting occurs with smooth crucibles. Experiments performed in graphitised crucibles are not, however, sufficiently documented to estimate the porosity and roughness of the coating.

– When de-wetting occurs, the structural quality of the crystals is improved (fewer dislocations, fewer grains) [1–3,10,18,19], probably owing to the disappearance of stresses exerted by the crucible on the crystal during cooling [41]. Moreover, bubbles are never observed in this case. Thermocapillary convection has never been reported in cases where free surfaces were due to de-wetting alone.

From this literature survey, it is clear that de-wetting is beneficial to crystal quality and that is appearance seems related to numerous factors: crystal growth angle, wetting angle, crucible roughness.

A directional solidification experiment was carried out in space (flight D1 of SPACELAB, experiment D1-GHF-03) using the InSb compound in a Bridgman set-up. The results of this experiment with regard to de-wetting (section 2) were used to propose an explanation of the phenomenon (section 3) which will be supported by ground experiments (section 4).

2. Experimental description

2.1. General

The parameters considered to be important to surface phenomena are as follows in decreasing order:

– The crucible material, which determines the liquid-crucible wetting angle and thus the geometry of the liquid zones. Vitreous fused silica crucibles were used in this experiment for reasons of chemical and thermal compatibility.

– The surface state of the crucible, which can modify the apparent wetting angle and may be a source of de-wetting.

– The residual atmosphere in the cartridge, likely to contaminate the surface, to change the surface tension, and to form oxide layers hindering capillary convection phenomena.

The existence and dimensions of a free meniscus which imposes internal pressures on the liquid, thereby leading to curvature of possible spontaneous free surfaces.

The internal shape of the crucible (sharp angles), which may also be source of de-wetting.

2.2. Experimental set-up

The directional solidification of InSb in a vitreous fused silica crucible is carried out in the SPACELAB Gradient Heating Facility (GHF) furnace. The experimental set-up is placed in a stainless steel cartridge and three cartridges are inserted together in the furnace. Two of these cartridges contain pure InSb samples and the third an InSb alloy containing 3000 ppm of tin (9×10^{19} atoms cm^{-3}).

Fig. 2 shows the experimental set-up in the cartridge. The relatively complex thermal system is described elsewhere [42]. In order to study the de-wetting phenomena, a variation in the surface area of the sample solidification front, $\Delta S = 100\%$, is imposed by an abrupt increase in diameter of the vitreous fused silica crucible, machined in the body of the crucible. The roughness of internal crucible surfaces is reduced by polishing (7 μm diamond powder). The crucible is subsequently cleaned with a mixture of HF, H_2O_2 and H_2O (in

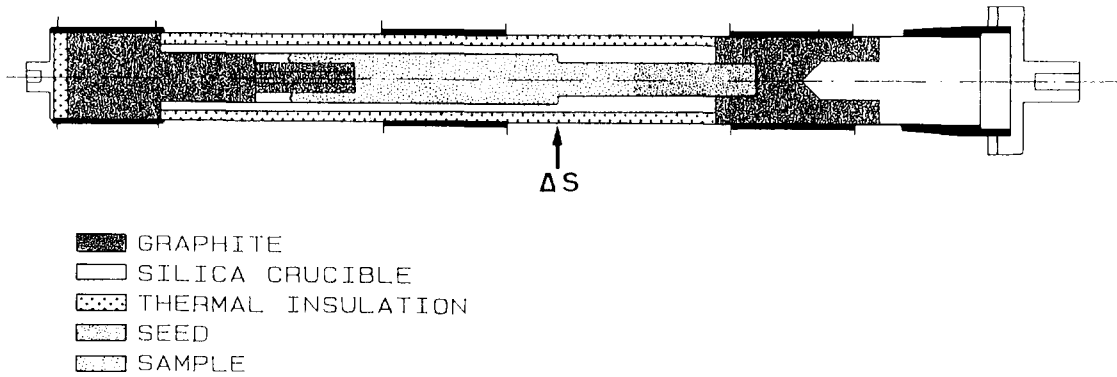

Fig. 2. Diagram of experimental cartridge.

the proportions 4 : 1 : 15) for 15 hours and then rinsed with ethanol before assembly. Roughness measurements were taken on reference crucibles. Some much rougher zones are left, in the same place for all the crucibles. To ensure liquid cohesion after fusing under microgravity conditions, the feed material and part of th seed are initially fused on the ground in order to weld the various parts of the sample.

A zirconium getter in the hot zone (650–750 °C) of the cartridge is used to eliminate oxygen and its derivatives (H_2O, CO_2) during the experiment while another getter is located in the cold zone (350–450 °C) to eliminate hydrogen. The various parts are cleaned and degassed before assembly and the entire assembly is then baked under vacuum (72 h, 400 °C, 10^{-4} Pa) before sealing the cartridge.

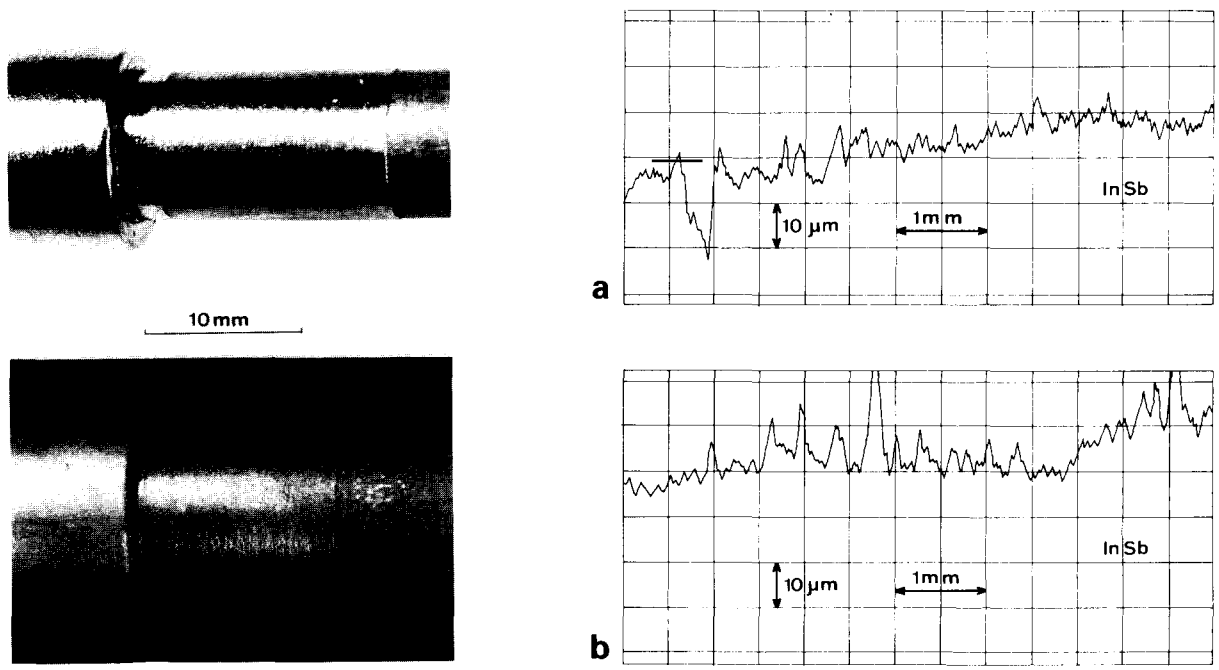

Fig. 3. Surface roughness of InSb crystals solidified in microgravity (a) and on the ground (b).

2.3. Experiment results

The residual vacuum in the cartridges, on the ground and in flight, after the experiment was found to be less than 0.5 Pa. This value, obtained one year after closure of the cartridges, shows that the getters correctly fulfilled their role. These residual gases were not chemically analysed.

After opening the cartridges, and in order to avoid breaking the samples, the silica crucible was dissolved in a 40% HF solution. Reference samples were checked to make sure that the sample surface remains intact after such treatment.

Crucible moulding was found to be satisfactory both on the ground and in flight (fig. 3). It was found, however, that in the ground reference experiment, the crucible roughness marks are clearer on the crystal surface; roughness measurements on the crystal surface give evidence of this difference in behaviour. De-wetting does not occur, nor is any accidental free surface generated at the sharp angles created by the abrupt variation in solidification front area.

Two wetting anomalies were observed in the microgravity experiment. First, a bubble-shaped cavity appeared at the seeding stage of the doped crystal. The disturbing effect of the free surface associated with this bubble on solute distribution has already been described and explained in terms of thermocapillary convection [25,26].

Secondly, a de-wetting zone is clearly visible on the doped sample surface (fig. 4) and, to a lesser extent, on the surface of the other, undoped space sample. This is a faceted surface, clearly characteristic of free growth without any contact with the crucible. The faceting can be explained by the fact that the sample was polycrystalline at this stage. Contrary to similar observations made by other authors (table 1), no ridges were visible and the zone is of limited extent. This de-wetting phenomenon takes place at the same point in the two crystals: this position corresponds to the crucible zone of greatest roughness.

As far as surface morphology is concerned, these experimental results support the literature survey results and confirm that the de-wetting phenomenon is directly related to crucible roughness.

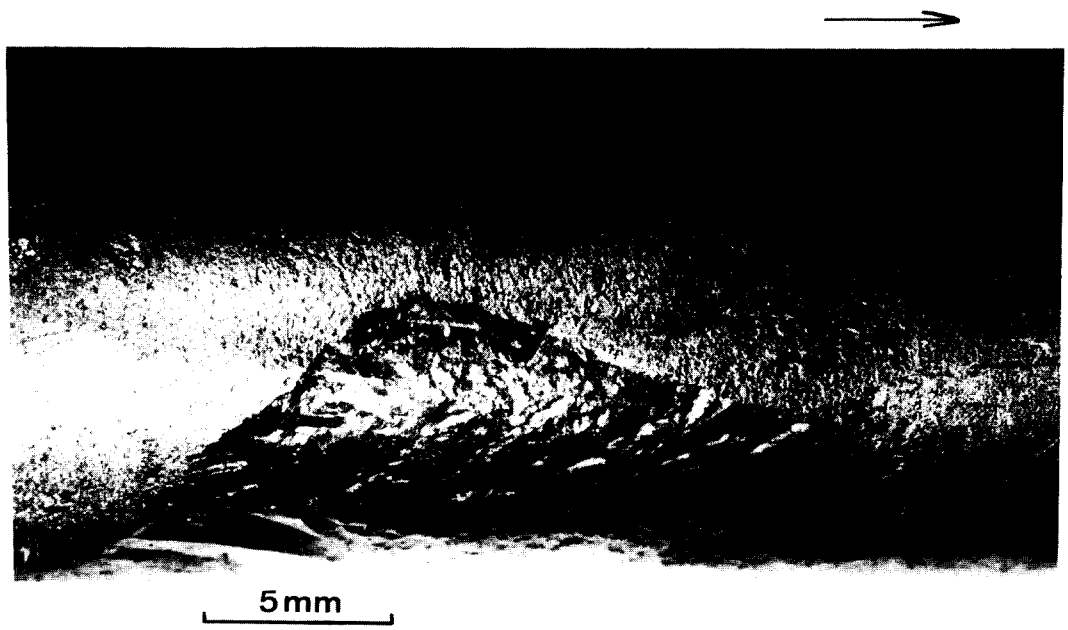

Fig. 4. De-wetting zone at the surface of an InSb crystal grown under microgravity conditions.

3. Theory

3.1. General

In view of the conclusions drawn in the preceding section, an explanation of the de-wetting phenomenon can be proposed, involving the classic concepts of roughness, wetting angle with notation θ_w, and growth angle, with notation α.

In that which follows, roughness at any point on the crucible is defined by the angle, r, between the tangent to the crucible at this point and the normal to the macroscopic surface of the crucible. The crucible is assumed to be macroscopically flat (i.e., at a scale where roughness is negligible). Finally, the phenomenon is simulated two-dimensionally, situated in a plane perpendicular to the macroscopic surface of the crucible and to the solidification interface.

A necessary condition for de-wetting is the initial achievement of composite wetting between the fused sample and the crucible. Explained by Johnson and Dettre [43], this type of wetting occurs when the crucible roughness and wetting angle are such that the liquid cannot penetrate to the bottom of the crucible cavities. Fig. 5 shows this type of wetting in simple cases where the roughness is simulated by a sine curve (rounded peaks) or by a saw-tooth curve (sharp peaks). The radius of curvature, R, of the liquid surface is given by the Laplace equation:

$$\Delta P = \sigma / R, \tag{1}$$

where ΔP = pressure difference and σ = surface tension.

On the ground, the liquid internal pressure is given by the hydrostatic pressure, which leads:

$$R = \sigma / \mu g h, \tag{2}$$

where μ = density of the liquid and h = height of liquid.

Under microgravity conditions, this pressure is negligible and a very wide radius of curvature will be assumed compared to the characteristic roughness dimensions ($R \approx \infty$). The liquid surface is then assumed to be flat.

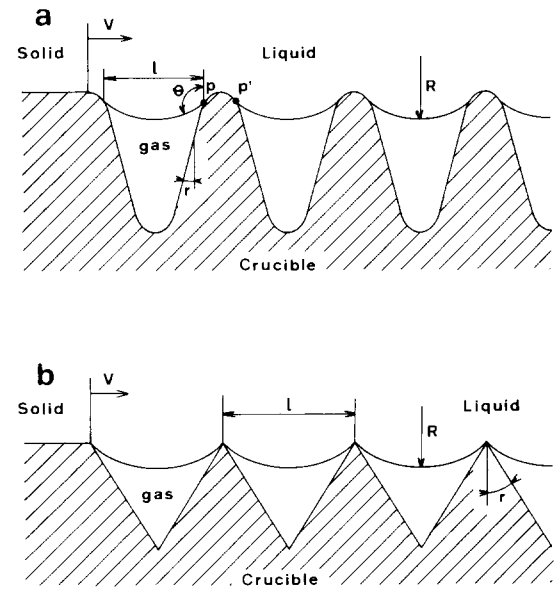

Fig. 5. Composite wetting in case of smooth roughness (a) and sharp roughness (b).

The mathematical condition for obtaining composite wetting is as follows (l is the distance separating the liquid–crucible contacts):

$$l/R \leq -2\cos(\theta_w - r). \tag{3}$$

Composite wetting is of course easier to obtain in microgravity where (with $R \approx \infty$) the condition (3) becomes:

$$r \leq \theta_w - \pi/2. \tag{3'}$$

3.2. Sharp peaks – analytical solutions

(a) *In microgravity.* Consideration is given to planar-front solidification of the liquid in contact with the sharp peaks of the crucible (fig. 6a) at time $t = 0$. The de-wetting phenomenon is explained in the following way: at time $t + \Delta t$, the solid–liquid interface has moved by Δx so that the angle between the solidified fraction and the liquid (adhering to the second roughness peak) is equal to the growth angle α (fig. 6b). Planar-front solidification continues in this way until the angle between the liquid and the second roughness peak reaches the wetting angle value θ_w (fig. 6c). At this stage, the liquid detaches from its anchor point

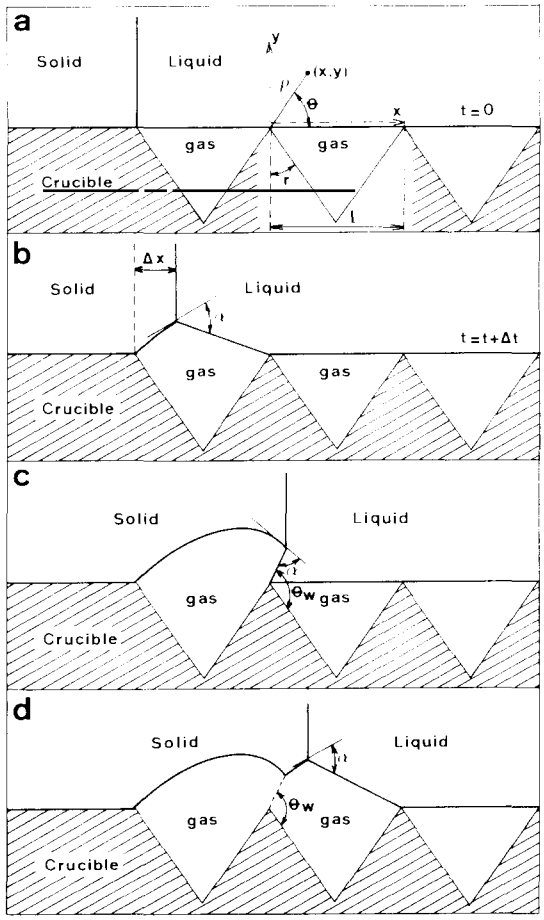

Fig. 6. Solidification process under microgravity conditions with sharp roughness crucible.

and the same procedure is repeated with the liquid attached to the next roughness peak (fig. 6d).

In this case, the problem is easily expressed in mathematical terms. The reference frame chosen is associated with the tip of the second roughness peak (fig. 6a). l represents the distance between two roughness peaks. The solidification trajectory is the curve that passes through all the components of the family of straight lines passing through the origin (0,0) according to the growth angle α.

This curve is called the α-trajectory and its analytical expression is obtained by the following property [44]: If the relation

$$f(x, y, y') = 0$$

is satisfied by a family of curves, the α-trajectory of this family of curves is given by:

$$f\left(x, y, \frac{\dot{y} - \tan\alpha}{1 + \dot{y}\tan\alpha}\right) = 0, \tag{4}$$

where $\dot{y}$ is the first derivative of the α-trajectory. In the case in question, the family of curves is: $y = ax$ (a variable) or $y' = a$, and $f(x, y, y') = y - y'x = 0$.

By substitution in (4) and after conversion into polar coordinates, the following differential equation is obtained:

$$\mathrm{d}\theta/\tan\alpha = \mathrm{d}\rho/\rho. \tag{5}$$

With initial conditions of $\rho = l$ and $\theta = \pi$, the following solution is obtained by integration of (5):

$$\rho = l\exp[(\theta - \pi)/\tan\alpha], \tag{6}$$

which is the equation of an infinite spiral. At the initial point, where the solid–liquid interface is still in contact with the crucible, α is undefined, and so this point is taken as an arbitrary initial boundary of the problem. In all cases, the detachment therefore takes place at a point such that $\theta = \theta_w + r - \pi/2$ (fig. 6c). To have the complete solidification profile, the system is shifted by one roughness peak and the detachment point is taken as new initial condition.

It can be concluded that, as soon as composite wetting conditions are satisfied, detachment takes place regardless of the growth angle.

(b) *On the ground.* The solidification principle and conditions (composite wetting, constant angle between solid and liquid, planar front) are identical to those described in (a) above in the case of microgravity. The only difference is that, due to gravity (hydrostatic pressure), the liquid radius of curvature is no longer negligible.

The solid shape calculation for this case (cf. appendix) also gives the equation of a spiral, similar to that obtained in microgravity:

$$\rho = 2R\exp[(\theta + k)/\tan\alpha]. \tag{7}$$

When ρ is small, the curvature of the liquid is no longer a significant factor and the liquid can be likened to a straight line.

Consequently, on the ground as under microgravity conditions, if composite wetting takes place then so does detachment.

3.3. Rounded peaks

System configuration is as represented on fig. 5a. To the authors' knowledge, there is no analytical solution for the solid shape calculation after planar-front solidification. In such a case, a numerical simulation of the phenomenon is used.

At time $t + \Delta t$, it is assumed that a fraction Δx of the liquid has solidified and a dichotomic method is used to determine the value Y of the solid–liquid–gas triple point located in Δx and satisfying the following solidification conditions:
- solid–liquid angle = growth angle α;
- the liquid is represented by a straight line (microgravity) or by a circle of radius R (ground);
- liquid–crucible angle = wetting angle θ_w.

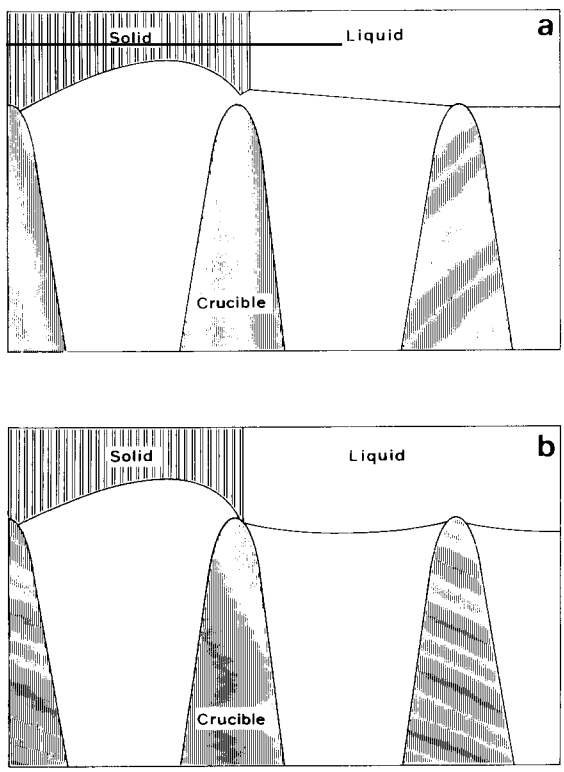

Fig. 7. Numerical simulation of de-wetting in case of smooth roughness: (a) in microgravity; (b) on ground.

The liquid contact point P on the roughness asperity naturally shifts to satisfy the above conditions.

The complete solidification profile is obtained by repeating the operation ad by checking at each step that the solid has not reached the next asperity and that the point P has not joined the point P′ (fig. 5a), as this would be the detachment condition.

The various simulations made by varying the four parameters (growth angle, wetting angle, roughness shape (sine curve) and radius of curvature R) showed that both cases exist: detachment or non-detachment, whether on the ground or in space. For certain specific values of the parameters (R, α, θ_w), detachment occurs in microgravity but not on the ground (fig. 7).

4. Experimental check-discussion

The results of section 3 show that de-wetting can be obtained in ground experiments. InSb solidification experiments according to the horizontal Bridgman technique were carried out under argon atmosphere in semi-cylindrical graphite crucibles machined in order to obtain the configuration shown in fig. 5b. The distance between roughness asperities varied from 300 to 500 μm, the roughness angle was 27.5° and the liquid height approximately 5 mm. In this way, composite wetting conditions were more than satisfied (eqs. (2) and (3), $\theta_w = 140°$, $\mu = 6.5 \times 10^3$ kg m^{-3}, $\sigma = 0.42$ N m^{-1}).

Fig. 8 shows the bottom surface of a polycrystal obtained in a graphite crucible (POCO AXM5-Q1) made with three different surface states: normal graphite, machined with a pitch of 300 μm, machined with a pitch of 500 μm. The striated and faceted zones correspond to de-wetting and, in particular, present similar ridges to those obtained in microgravity by a number of different authors. Facets are due to dendritic crystal growth as a result of a high level of supercooling. These facets are much less marked in the unmachined zone of the crucible although slight de-wetting is still apparent.

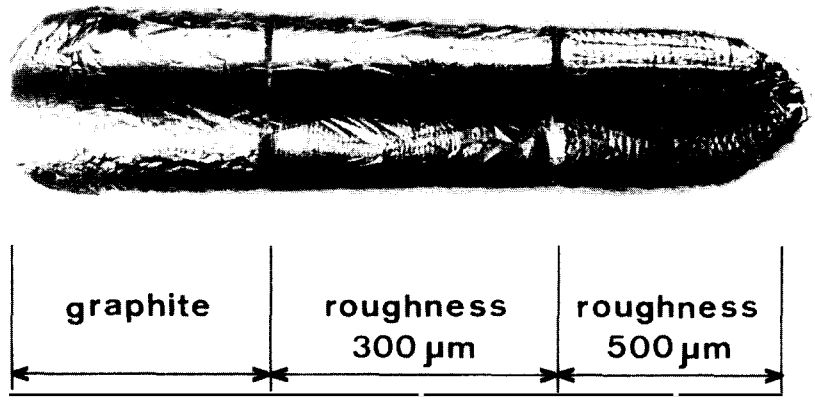

Fig. 8. Bottom surface of an InSb sample solidified in a graphite crucible with variable machined roughness ($\alpha \approx 30°$, $\theta_w > 140°$, $r = 28°$).

The experiment represented on fig. 9 involved a crucible with only a few roughness zones machined in the graphite (1346 PT Carbon Lorraine). Crystal growth was from a seed with a drawing rate of 3.4 μm/s. The surface state of the crystal is comparable to that of the crucible, except in the machined roughness zones where free-surface growth has clearly taken place.

Fig. 9 gives a detail of de-wetting in the section where the crucible had three different roughness zones. Three-dimensional roughness-meter analyses of the bottom part of the crystal in this section clearly show that the calculations presented in section 3 above can be applied to describe solidification under these conditions.

The results of the microgravity experiment described in section 2, for which de-wetting occurs only when a critical roughness value is reached, are compatible with the analysis presented. From measurements of the surface state, the mean roughness angle can be estimated at about 45°. Under these conditions, an InSb wetting angle on silica of 135° must be assumed in order to reach the composite wetting condition (3′). For this experiment, hydrostatic pressure prevents any composite wetting during solidification on the ground.

Composite wetting and then solidification lead to the existence of free liquid surfaces which can give rise to capillary convection. The surface liquid velocity is calculated from the specific Reynolds number [45]:

$$\mathrm{Re}_M = \frac{d\sigma}{dT}\frac{Gl^2}{\mu\nu^2}, \tag{8}$$

where $d\sigma/dT$ = variation of surface tension with T, G = thermal gradient, μ = liquid specific weight, and ν = viscosity.

In case of composite wetting, since the characteristic dimension l is equal to the distance between asperities, which is of the order of a few tens of micrometers, the Reynolds number is reduced by several orders of magnitude with respect to cases for which capillary convection impedes the solidification process. If de-wetting propagates over several asperities ahead of the interface, necking is obtained (cf. section 1). However, the wetting angle is then replaced by a preceding angle $\theta_w - r$ and necking is more difficult to obtain since the condition for its appearance becomes:

$$\theta_w + \alpha - r \geq \pi.$$

In general, the surfaces of crystals with de-wetting have ridges, small asperities, etc. Lendvay et al. [20] explain this phenomenon by the emergence of dislocations which disturb the local crystal growth conditions. Ukanwa [11] puts forward the theory of very localised thermal disturbances. While such explanations cannot be rejected out of

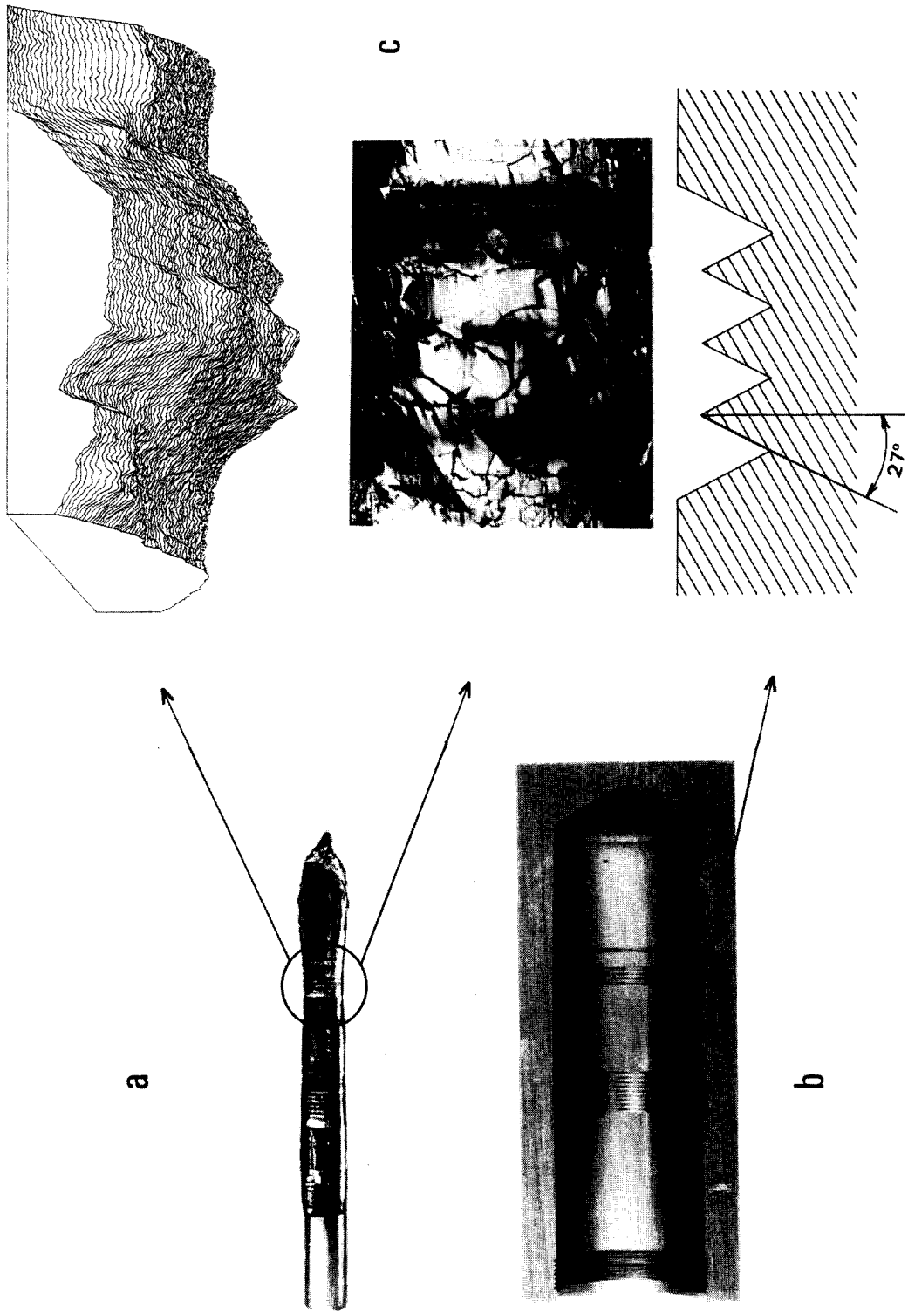

Fig. 9. Crystallization of InSb crystal in graphite crucible of variable roughness ($\alpha \approx 30°$, $\theta_w \geq 140°$, $r = 28°$): (a) crystal; (b) crucible: (c) local surface roughness.

hand, it is possible that the local roughness conditions could cause this types of asperity. The few cases of de-wetting in smooth crucibles (cf. table 1) are not taken into account by the proposed explanation. Additional experimentation is therefore needed to clarify this point.

The mechanism proposed here also provides a solution for understanding why crystals are easily removed from the moulds when grown on the ground in a sand blasted crucible. Both Kang et al. [46] and Cho et al. [47], showed that the detachment of a GaAs alloy grown by horizontal Bridgman solidification from silica crucibles was obtained only if the particle size of the powder used for sand blasting was less than about 50 μm. There is reason to believe that composite wetting appears only from the corresponding roughness value and only then would de-wetting become possible.

6. Conclusions

A mechanism involving the semiconductor growth angle, its wetting angle on the crucible and the crucible surface roughness is proposed for explaining de-wetting of the crucible by the liquid during Bridgman growth in microgravity. Before this phenomenon can occur, composite wetting between liquid and crucible must first be obtained. It is thus shown that detachment takes place just in front of the solidification interface, over a characteristic dimension equal to the dimension of the roughness asperities. In the event of sharp peaks, this condition is sufficient both on the ground and in microgravity.

The same mechanism can be used to explain detachment of the crystal obtained on the ground by sand-blasting the crucible with a sufficiently fine powder.

The de-wetting phenomenon is beneficial for crystal quality from several standpoints:

– The necessary composite wetting condition leads to a reduction in contact area between crucible and liquid and thus to a reduction in chemical pollution of the liquid.

Composite wetting, coupled with the absence of contact between crystal and crucible, creates thermal resistance between the sample and the crucible, thus reducing the radial heat flux that generally exists at the solid–liquid interface owing to thermal conductivity differences. The flatness of the solid-liquid interface can thus be improved.

– The stresses generally exerted on the crystal by the crucible, as a result of differential expansion during cooling, no longer exist. Experience has shown that the quality of crystals is generally better, in terms of the number of dislocations, in the case of de-wetting in microgravity compared to the same experiments performed on the ground without de-wetting.

In all cases of Bridgman semiconductor crystal growth, it is recommended to use crucibles with strictly controlled roughness characteristics, whether for experiments on the ground or in microgravity. Moreover, for each material to be solidified, it is important to choose a crucible material not only in terms of its physico-chemical properties, but also in term of its wetting properties with respect to the liquid semiconductor. In order to be able to control de-wetting and so that it can be repeated experimentally, it is essential to carry out systematic measurements of growth and wetting angles of semiconductors and their crucibles.

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Appendix. Calculation of solid shape in cases of hydrostatic pressure

Let the initial frame of reference be as shown in fig. 10. The α-trajectory with respect to circles of radius R passing through the origin must be determined.

Consider a point $P(\rho, \theta)$ common to the circle C and to the α-trajectory T. The derivative of the

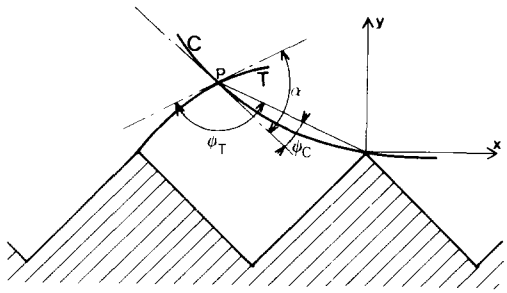

Fig. 10. Frame of reference and definition of angles for calculations in appendix.

circle in this point is given the notation ρ' and that of the α-trajectory $\dot{\rho}$.

We thus have:

$$\pi - \alpha = \psi_C - \psi_T,$$

where

$$\tan \psi_T = \rho/\dot{\rho}, \quad \tan \psi_C = \rho/\rho'.$$

Let

$$\tan \alpha = \tan(\psi_T - \psi_C). \tag{A.1}$$

The expression for ρ' is therefore:

$$\rho' = \frac{\rho^2 \tan \alpha + \rho\dot{\rho}}{\rho - \dot{\rho} \tan \alpha}. \tag{A.2}$$

A property analogous to that given in microgravity is thus obtained: if ρ' satisfies $f(\rho, \theta, \rho') = 0$, than $\dot{\rho}$ satisfies:

$$f\left(\rho, \theta, \frac{\rho^2 \tan \alpha + \rho\dot{\rho}}{\rho - \dot{\rho} \tan \alpha}\right) = 0. \tag{A.3}$$

This property is used to find the α-trajectory with respect to the family of circles passing through the point (0, 0) and of radius R. The equation of such circles is of the following form:

$$\rho = 2R \cos(\theta - \beta) \rightarrow \rho' = \left(4R^2 - \rho^2\right)^{1/2},$$

where (R, ρ) are the coordinates of the centre of the circle at each instant.

By applying eq. (A.3), the differential equation of the trajectory is obtained:

$$\rho^2 \tan \alpha + \rho\dot{\rho} = \left(4R^2 - \rho^2\right)^{1/2}(\rho - \dot{\rho} \tan \alpha). \tag{A.4}$$

By substituting the variable $U = (4R^2/\rho^2 - 1)^{1/2}$, the equation becomes

$$-\frac{1}{U^2 + 1} - \frac{\tan \alpha}{U - \tan \alpha} = \mathrm{d}\theta. \tag{A.5}$$

It should be noted that $U = \tan \alpha$ is a particular solution corresponding to the circle with centre at (0,0) and or radius $2R \cos \alpha$.

Integration of eq. (A.5), with initial conditions $\rho = l$ and $\theta = \pi$, gives the final equation:

$$-\arctan U + \arctan U_0 - \tan \alpha \log\left(\frac{U - \tan \alpha}{U_0 - \tan \alpha}\right) = \theta - \pi, \tag{A.6}$$

where $U_0 = (4R^2/l^2 - 1)^{1/2}$.

This equation can be expressed in the following form:

$$-\arctan U - \tan \alpha \log(U - \tan \alpha) = \theta + \text{constant}. \tag{A.7}$$

It now remains to be seen whether this curve touches the asperity before $\theta = \theta_w + r - \pi/2$.

If it is assumed that the solidification curve approaches the asperity, then ρ becomes small and U thus becomes great. Under these conditions, the following approximations can be made:

$-\arctan U \approx -\pi/2 \approx$ constant,

$\tan \alpha$ is negligible compared to U,

$U \approx 2R/\rho$.

Eq. (A.7) can then be written:

$$\log(\rho/2R) = \frac{\theta + k}{\tan \alpha} \quad (k = \text{constant})$$

or

$$\rho = 2R \exp\left(\frac{\theta + k}{\tan \alpha}\right). \tag{A.8}$$

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